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Robust uncertainty principles: exact signal reconstruction from highly incomplete frequency information

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Domain Signal processing, statistics & applied mathematics

Abstract

This paper considers the model problem of reconstructing an object from incomplete frequency samples. Consider a discrete-time signal f∈C^N and a randomly chosen set of frequencies Ω. Is it possible to reconstruct f from the partial knowledge of its Fourier coefficients on the set Ω? A typical result of this paper is as follows. Suppose that f is a superposition of |T| spikes f(t)=Σ_τ∈Tf(τ)δ(t-τ) obeying |T|≤C_M·(log N)^-1 · |Ω| for some constant C_M>0. We do not know the locations of the spikes nor their amplitudes. Then with probability at least 1-O(N^-M), f can be reconstructed exactly as the solution to the ℓ_1 minimization problem. In short, exact recovery may be obtained by solving a convex optimization problem. We give numerical values for C_M which depend on the desired probability of success. Our result may be interpreted as a novel kind of nonlinear sampling theorem. In effect, it says that any signal made out of |T| spikes may be recovered by convex programming from almost every set of frequencies of size O(|T|·logN). Moreover, this is nearly optimal in the sense that any method succeeding with probability 1-O(N^-M) would in general require a number of frequency samples at least proportional to |T|·logN. The methodology extends to a variety of other situations and higher dimensions. For example, we show how one can reconstruct a piecewise constant (one- or two-dimensional) object from incomplete frequency samples - provided that the number of jumps (discontinuities) obeys the condition above - by minimizing other convex functionals such as the total variation of f.

Real-World Impact & Application

Revolutionizing Medical MRI & Medical Scanners: Candes, Romberg, and Tao proved the mathematical foundation of Compressed Sensing: sparse physical signals can be reconstructed with 100% exactness from a fraction of traditional sensor measurements. This breakthrough directly enabled modern hospital MRI scanners to reduce patient imaging times from 45 minutes to under 5 minutes without losing diagnostic detail, dramatically improving patient comfort and clinical throughput.